

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
First , let's calculate the heat required to convert the liter of H₂O at 30°C to a liter of water at 60°C . The heat required for this process can be calculated using the specific heat capacity of water and the change in temperature . For the conversion and the heating process together , the total heat needed can be calculated using the formula :
$$Q = m \times c_p \times \Delta T$$
where m is the mass , c_p is the specific heat capacity , and ΔT is the change in temperature . We need to convert liters to grams since the specific heat is given in cal/g°C . Since 1 liter of H₂O at 30°C converts to 1000 g (since the density of water is approximately 1 g/cm³), the mass m is approximately 1000 g .

- For the conversion process :
- Initial temperature $T_i = 30^{\circ}C$
 - Final temperature $T_f = 60^{\circ}C$
 - Temperature change $\Delta T = 60 - 30 = 30$

So , the heat required for the conversion is :
$$Q_{\text{conversion}} = 1000 \text{ g} \times 1 \text{ cal/g}^{\circ}C \times 30 \text{ }^{\circ}C = 30000 \text{ cal}$$

- For the heating process , we have :
- Mass $m = 1000 \text{ g}$
 - Specific heat $c_p = 1 \text{ cal/g}^{\circ}C$
 - Temperature change $\Delta T = 60 - 30 = 30 \text{ }^{\circ}C$

So , the heat required for heating is :
$$Q_{\text{heating}} = 1000 \text{ g} \times 1 \text{ cal/g}^{\circ}C \times 30 \text{ }^{\circ}C = 30000 \text{ cal}$$

So , the total heat Q_{total} needed is :
$$Q_{\text{total}} = 30000 \text{ cal} + 30000 \text{ cal} = 60000 \text{ cal}$$

This heat Q_{total} is approximately $\frac{60000}{1000} = 60 \text{ kJ}$ and converting to kcal (1 kJ = 0.2390057361 kJ), we have $60 \times 0.2390057361 = 14.340344156 \text{ kcal} \approx 14.34 \text{ kcal}$.

- Next , for the heating of a 1 kg block of aluminum , the heat required is :
- $$Q = m \times c_p \times \Delta T$$
- Mass $m = 1 \text{ kg} = 1000 \text{ g}$
 - Specific heat $c_p = 0.215 \text{ cal/g}^{\circ}C$
 - Temperature change $\Delta T = 60 - 30 = 30 \text{ }^{\circ}C$